STA301 - Statistics and Probability (QUIZ # 03)

Question # 6 of 10 (Start time: 12:33:36 PM, 08 January 2026)

Mean of binomial distribution is:

Select the correct option

○	n
○	p
○	n/p
●	np

STA301 - Statistics and Probability (QUIZ # 03)

Question # 5 of 10 (Start time: 12:33:24 PM, 08 January 2026)

In sampling from a large population with δ= 60. The size of the sample used is 100, the value of the standard error will be:

Select the correct option

- 4
- 6
- 15
- 42

STA301 - Statistics and Probability (QUIZ # 03) Quiz Start Time: 12:32 PM, 08 January 2026

Question # 3 of 10 (Start time: 12:33:02 PM, 08 January 2026) **Total Marks:** 1

Consider a large population with a mean of 160 and a standard deviation of 20. A random sample of size 64 is taken from this population. What is the standard deviation of the sample mean?

Select the correct option

- 3.568
- 3.125
- 2.500
- 1.654

STA301 – Statistics and Probability (QUIZ # 03)

Question # 4 of 10 (Start time: 12:33:14 PM, 08 January 2026)

If X and Y are independent, then Cov (X, Y) is equal to:

Select the correct option

○	1
○	None
●	0
○	0 and 1

STA301 - Statistics and Probability (QUIZ # 03)

Question # 2 of 10 (Start time: 12:32:52 PM, 08 January 2026)

An estimator is always a:

Select the correct option

●	Statistic and Random variable
○	Random variable
○	Parameter
○	Statistic

STA301 - Statistics and Probability (QUIZ # 03)

Question # 1 of 10 (Start time: 12:32:39 PM, 08 January 2026)

The number of trials in hypergeometric distribution is:

Select the correct option

◉	Fixed
○	Small
○	Not fixed
○	Large

STA301 - Statistics and Probability (QUIZ # 03)

Question # 10 of 10 (Start time: 12:34:31 PM, 08 January 2026)

$According\ to\ emperical\ rule,\ how\ much\ data\ lies\ between\ \mu - 3\sigma\ and\ \mu + 3\sigma\ ?$

Select the correct option

○	95.44%
○	68.26%
●	99.73%
○	50%

Click

STA301 - Statistics and Probability (QUIZ # 03)

Question # 9 of 10 (Start time: 12:34:11 PM, 08 January 2026)

From joint distribution function f(X, Y), we can get marginal probabilities function, but vise versa is true when

Select the correct option

●	X and Y are independent
○	X and Y are highly correlated
○	X and Y are dependent
○	X and Y are symmetrical

STA301 - Statistics and Probability (QUIZ # 03)

Question # 7 of 10 (Start time: 12:33:47 PM, 08 January 2026)

In a normal distribution how much area lies between $\mu \pm 3\sigma$

Select the correct option

◉	99.73%
○	70%
○	68%
○	89%

STA301 - Statistics and Probability (QUIZ # 03) Quiz St

Question # 6 of 10 (Start time: 12:29:20 PM, 08 January 2026)

Which of the following statements is correct?

Select the correct option

○	A point estimate is an unbiased estimator if its standard deviation is the same as the actual value of the population standard deviation
○	All of the above statements are correct
◉	A point estimate is an estimate of the range of a population parameter
○	A point estimate is a single value estimate of the value of a population parameter

STA301 - Statistics and Probability (QUIZ # 03)

Question # 8 of 10 (Start time: 12:33:59 PM, 08 January 2026)

Which of the following is NOT a requirement of a binomial distribution?

Select the correct option

○	Only two possible outcomes
○	A fixed number of trials
○	A constant probability of success
●	Equally likely outcomes

STA301 - Statistics and Probability (QUIZ # 03)

Question # 5 of 10 (Start time: 12:29:11 PM, 08 January 2026)

The probability that a continuous random variable X will assume any specific value is

Select the correct option

- (●) 0
- ( ) 1
- ( ) 0.50
- ( ) any value between -0.5 and 0.5

STA301 - Statistics and Probability (QUIZ # 03) Quiz Start Time: 12:28 PM, 08 January 2026

Question # 4 of 10 (Start time: 12:28:59 PM, 08 January 2026) **Total Marks:** 1

Suppose we have a population of size N = 5 i.e. 0,11,20,23,45 and if we draw all possible samples of size n = 2 with replacement then total possible samples will be:

Select the correct option

- ( ) 10
- (●) 25
- ( ) 35
- ( ) 50

Question # 3 of 10 (Start time: 12:28:49 PM, 08 January 2026)

The hyper geometric probability distribution is appropriate when a random sample of size n is drawn without replacement from a _____ population of N units:

Select the correct option

- ○ total
- ○ infinite
- ◉ finite
- ○ All

STA301 - Statistics and Probability (QUIZ # 03)

Question # 2 of 10 (Start time: 12:28:38 PM, 08 January 2026)

Which type of data is used in the binomial distribution?

Select the correct option

○	Continues data only
●	Discrete data only
○	Qualitative data only
○	Both discrete and continuous

STA301 - Statistics and Probability (QUIZ # 03)

Question # 1 of 10 (Start time: 12:28:24 PM, 08 January 2026)

What should be the type of both variables in joint probability distribution of X and Y?

Select the correct option

○	One continuous and other discrete
◉	Either both continuous or both discrete
○	Both continuous
○	Both discrete

STA301 - Statistics and Probability (QUIZ # 03)

Question # 10 of 10 (Start time: 12:30:11 PM, 08 January 2026)

The marginal p.d.f. of any continuous variables is obtained by _____ the other variable from the joint p.d.f.

Select the correct option

- ○ Averaging
- ○ Multiplying
- ○ Dividing
- ◉ Integrating

STA301 - Statistics and Probability (QUIZ # 03) Quiz Start

Question # 8 of 10 (Start time: 12:29:41 PM, 08 January 2026)

If a statistic used as an estimator, has its expected value equal to the true value of the population parameter being estimated then it is called .....

Select the correct option

- ○ Consistent
- ○ Efficient
- ○ Sufficient
- ◉ Unbiased

Click to Save

STA301 - Statistics and Probability (QUIZ # 03)

Question # 9 of 10 (Start time: 12:29:50 PM, 08 January 2026)

If X and Y are independent random variables, then Cov (X, Y) =

Select the correct option

◉	0
○	1
○	0.5
○	2

STA301 - Statistics and Probability (QUIZ # 03)

Question # 7 of 10 (Start time: 12:29:30 PM, 08 January 2026)

The normal curve is:

Select the correct option

○	Bimodal.
○	Multimodal.
◉	Symmetrical/Bell shaped
○	Asymmetrical